P24.1 For a uniform electric field passing through a plane surface, $\Phi_E = \vec{\mathbf{E}} \cdot \vec{\mathbf{A}} = EA\cos\theta$, where θ is the angle between the electric field and the normal to the surface.

(a) The electric field is perpendicular to the surface, so $\theta = 0°$:

$$\Phi_E = \left(6.20 \times 10^5 \text{ N/C}\right)\left(3.20 \text{ m}^2\right)\cos 0°$$

$$\Phi_E = \boxed{1.98 \times 10^6 \text{ N} \cdot \text{m}^2/\text{C}}$$

(b) The electric field is parallel to the surface: $\theta = 90°$, so $\cos\theta = 0$, and the flux is $\boxed{\text{zero.}}$

P24.3 For a uniform field the flux is $\Phi = \vec{\mathbf{E}} \cdot \vec{\mathbf{A}} = EA\cos\theta$.

The maximum value of the flux occurs when $\theta = 0$, or when the field is in the same direction as the area vector, which is defined as having the direction of the perpendicular to the area. Therefore, we can calculate the field strength at this point as

$$E = \frac{\Phi_{\max}}{A} = \frac{\Phi_{\max}}{\pi r^2}$$

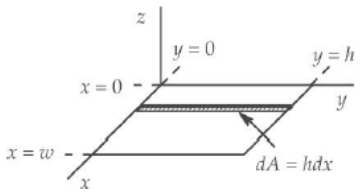
$$E = \frac{5.20 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}}{\pi(0.200 \text{ m})^2} = 4.14 \times 10^6 \text{ N/C} = \boxed{4.14 \text{ MN/C}}$$

P24.6 We are given an electric field in the general form

$$\vec{\mathbf{E}} = ay\hat{\mathbf{i}} + bz\hat{\mathbf{j}} + cx\hat{\mathbf{k}}$$

In the *xy* plane, $z = 0$ so that the electric field reduces to

$$\vec{\mathbf{E}} = ay\hat{\mathbf{i}} + cx\hat{\mathbf{k}}$$



ANS. FIG. P24.6

To obtain the flux, we integrate (see ANS. FIG. P24.6 for the definition of *dA*):

$$\Phi_E = \int \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \int \left(ay\hat{\mathbf{i}} + cx\hat{\mathbf{k}}\right) \cdot \hat{\mathbf{k}}\, dA$$

$$\Phi_E = ch \int_{x=0}^{w} x\, dx = ch\frac{x^2}{2}\bigg|_{x=0}^{w} = \boxed{\frac{chw^2}{2}}$$

Where the $\hat{\mathbf{k}}$ term was eliminated since $\hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = 0$.

P24.7 The electric flux through the hole is given by Gauss's Law (Equation 24.6) as

$$\Phi_{E,\text{ hole}} = \vec{\mathbf{E}} \cdot \vec{\mathbf{A}}_{\text{hole}} = \left(\frac{k_e Q}{R^2}\right)\left(\pi r^2\right)$$

$$= \left(\frac{\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left(10.0\times10^{-6}\ \text{C}\right)}{\left(0.100\ \text{m}\right)^2}\right) \times\pi\left(1.00\times10^{-3}\ \text{m}\right)^2$$

$$= \boxed{28.2\ \text{N}\cdot\text{m}^2/\text{C}}$$

P24.8 The gaussian surface encloses the +1.00-nC and –3.00-nC charges, but not the +2.00-nC charge. The electric flux is therefore

$$\Phi_E = \frac{q_{\text{in}}}{\epsilon_0} = \frac{\left(1.00\times10^{-9}\ \text{C} - 3.00\times10^{-9}\ \text{C}\right)}{8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2} = \boxed{-226\ \text{N}\cdot\text{m}^2/\text{C}}$$

P24.10 (a) From $E = \frac{k_e Q}{r^2}$,

$$Q = \frac{Er^2}{k_e} = \frac{\left(8.90\times10^{2}\,\text{N/C}\right)\left(0.750\ \text{m}\right)^2}{\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)} = 5.57\times10^{-8}\ \text{C}$$

But Q is negative since $\vec{\mathbf{E}}$ points inward, so

$$Q = -5.57\times10^{-8}\ \text{C} = \boxed{-55.7\ \text{nC}}$$

(b) The | negative | charge has a | spherically symmetric | charge distribution, concentric with the spherical shell.

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P24.13 Consider as a gaussian surface a box with horizontal area A, lying between 500 and 600 m elevation. From Gauss's Law,

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q}{\epsilon_0}:$$

$$(+120\ \text{N/C})A + (-100\ \text{N/C})A = \frac{\rho A(100\ \text{m})}{\epsilon_0}$$

$$\rho = \frac{(20.0\ \text{N/C})\left(8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2\right)}{100\ \text{m}} = \boxed{1.77\times10^{-12}\ \text{C/m}^3}$$

The charge is $\boxed{\text{positive}}$, to produce the net outward flux of electric field.

P24.17 (a) If $R \le d$, the sphere encloses no charge and $\Phi_E = \frac{q_{\text{in}}}{\epsilon_0} = \boxed{0}$.

(b) If $R > d$, the length of line falling within the sphere is $2\sqrt{R^2 - d^2}$

so $\Phi_E = \boxed{\dfrac{2\lambda\sqrt{R^2 - d^2}}{\epsilon_0}}$

P24.19 The total charge is $Q - 6|q|$. The total outward flux from the cube is $\frac{Q - 6|q|}{\epsilon_0}$, of which one-sixth goes through each face:

$$(\Phi_E)_{\text{one face}} = \frac{Q - 6|q|}{6\,\epsilon_0}$$

$$(\Phi_E)_{\text{one face}} = \frac{Q - 6|q|}{6\,\epsilon_0} = \frac{(5.00 - 6.00)\times10^{-6}\ \text{C}\cdot\text{N}\cdot\text{m}^2}{6\times8.85\times10^{-12}\ \text{C}^2}$$

$$= \boxed{-18.8\ \text{kN}\cdot\text{m}^2/\text{C}}$$

P24.24 Note that the electric field in each case is directed radially inward, toward the filament. We use $E = \dfrac{2k_e\lambda}{r}$ and substitute numerical values.

(a) At r = 10.0 cm = 0.100 m,

$$E = \frac{2k_e\lambda}{r} = \frac{2\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left(90.0\times10^{-6}\ \text{C/m}\right)}{0.100\ \text{m}}$$

$$= \boxed{16.2\ \text{MN/C}}$$

P24.25 The charge per unit area of the plastic sheet must be sufficiently large to result in an upward electric force on the Styrofoam that cancels the downward gravitational force:

$$mg = qE = q\left(\frac{\sigma}{2\epsilon_0}\right) = q\left(\frac{Q/A}{2\epsilon_0}\right)$$

Solving for the charge per unit area gives

$$\frac{Q}{A} = \frac{2\epsilon_0 mg}{q}$$

$$= \frac{2\left(8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2\right)\left(10.0\times10^{-3}\ \text{kg}\right)\left(9.80\ \text{m/s}^2\right)}{\left|-0.700\times10^{-6}\ \text{C}\right|}$$

$$= \boxed{2.48\ \mu\text{C/m}^2}$$

P24.27 For a large uniformly charged sheet, $\vec{\text{E}}$ will be perpendicular to the sheet, and will have a magnitude of

$$E = \frac{\sigma}{2\epsilon_0} = 2\pi k_e\sigma$$

$$= (2\pi)\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left(9.00\times10^{-6}\ \text{C/m}^2\right)$$

so $$\vec{\text{E}} = 5.08\times10^{5}\ \text{N/C}\ \hat{\text{j}}$$

P24.29 (a) Consider the spherical symmetry of the situation. A gaussian sphere concentric wth the shell, with radius 10.0 cm, encloses 0 charge. Then at the surface of this sphere, inside the charged shell, we have $\vec{\mathbf{E}} = \boxed{0}$.

(b) For a gaussian sphere of radius 20.0 cm, we apply $\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q_{in}}{\epsilon_0}$.

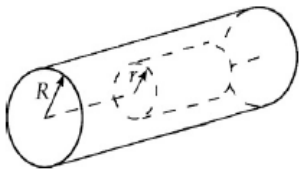
The field is radially outward, and $4\pi r^2 E = q/\epsilon_0$:

$$E = \frac{k_e q}{r^2} = \frac{\left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2\right)\left(32.0 \times 10^{-6} \text{ C}\right)}{(0.200 \text{ m})^2}$$

$$= 7.19 \times 10^6 \text{ N/C}$$

so $\vec{\mathbf{E}} = \boxed{7.19 \text{ MN/C radially outward}}$

P24.33 If ρ is positive, the field must be radially outward. Choose as the gaussian surface a cylinder of length L and radius r, contained inside the charged rod. Its volume is $\pi r^2 L$ and it encloses charge $\rho \pi r^2 L$. Because the charge distribution is long, no electric flux passes through the circular end caps; $\vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = EdA\cos 90.0° = 0$. The curved surface has $\vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = EdA\cos 0°$, and E must be the same strength everywhere over the curved surface.



ANS. FIG. P24.33

Gauss's law, $\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q}{\epsilon_0}$, becomes $E \int_{\substack{\text{Curved} \\ \text{Surface}}} dA = \frac{\rho \pi r^2 L}{\epsilon_0}$.

Now the lateral surface area of the cylinder is $2\pi rL$:

$$E(2\pi r)L = \frac{\rho \pi r^2 L}{\epsilon_0}$$

Thus, $\vec{\mathbf{E}} = \boxed{\frac{\rho r}{2\epsilon_0} \text{ radially away from the cylinder axis}}$.

P24.37 $\oint E\,dA = E(2\pi rl) = \frac{q_{\text{in}}}{\epsilon_0}$ $E = \frac{q_{\text{in}}/l}{2\pi\,\epsilon_0\,r} = \frac{\lambda}{2\pi\,\epsilon_0\,r}$ for the field outside the metal rod.

(a) At r = 3.00 cm, $\vec{\mathrm{E}} = \boxed{0}$

(b) At r = 10.0 cm,

$$\vec{\mathrm{E}} = \frac{30.0\times10^{-9}\ \mathrm{C}}{2\pi\left(8.85\times10^{-12}\ \mathrm{C^2/N\cdot m^2}\right)(0.100\ \mathrm{m})}$$

$$= \boxed{5\,400\ \mathrm{N/C,\ outward}}$$

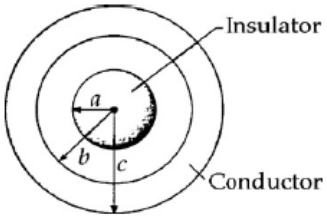
(c) At r = 100 cm,

$$\vec{\mathrm{E}} = \frac{30.0\times10^{-9}\ \mathrm{C}}{2\pi\left(8.85\times10^{-12}\ \mathrm{C^2/N\cdot m^2}\right)(1.00\ \mathrm{m})}$$

$$= \boxed{540\ \mathrm{N/C,\ outward}}$$

P24.54 Choose as each gaussian surface a concentric sphere of radius r. The electric field will be perpendicular to its surface, and will be uniform in strength over its surface. The density of charge in the insulating sphere is

$$\rho = Q / \left(\tfrac{4}{3}\pi a^3\right)$$



ANS. FIG. P24.54

(a) The sphere of radius $r < a$ encloses charge

$$q_{\text{in}} = \rho\left(\frac{4}{3}\pi r^3\right) = \left(\frac{Q}{\frac{4}{3}\pi R^3}\right)\left(\frac{4}{3}\pi r^3\right) = \boxed{Q\left(\frac{r}{R}\right)^3}$$

(b) Applying Gauss's law to this sphere reveals the magnitude of the field at its surface.

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q_{\text{in}}}{\epsilon_0}$$

$$E\left(4\pi r^2\right) = \frac{Q}{\epsilon_0}\left(\frac{r}{a}\right)^3 \rightarrow E = \frac{1}{4\pi\,\epsilon_0}\frac{Qr}{a^3} = \boxed{k_e\frac{Qr}{a^3}}$$

(c) For a sphere of radius r with $a < r < b$, the whole insulating sphere is enclosed, so the charge within is Q: $q_{\text{in}} = \boxed{Q}$.

(d) Gauss's law for this sphere becomes:

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q_{\text{in}}}{\epsilon_0}$$

$$E\left(4\pi r^2\right) = \frac{Q}{\epsilon_0} \rightarrow E = \frac{1}{4\pi\,\epsilon_0}\frac{Q}{r^2} = \boxed{k_e\frac{Q}{r^2}}$$

(e) For $b \le r \le c$, $\boxed{E = 0}$ because there is no electric field inside a conductor.

(f) For $b \le r \le c$, we know $E = 0$. Assume the inner surface of the hollow sphere holds charge Q_{inner}. By Gauss's law,

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q_{\text{in}}}{\epsilon_0}$$

$$0 = \frac{Q + Q_{\text{inner}}}{\epsilon_0} \rightarrow Q_{\text{inner}} = \boxed{-Q}$$

(g) The total charge on the hollow sphere is zero; therefore, charge on the outer surface is opposite to that on the inner surface:

$$Q_{outer} = -Q_{inner} = \boxed{+Q}$$

(h) A surface of area A holding charge Q has surface charge $\sigma = q/A$. The solid, insulating sphere has small surface charge because its total charge Q is uniformly distributed throughout its volume. The inner surface of radius b has smaller surface area, and therefore larger surface charge, than the outer surface of radius c.